

Analog input (UG-637 Fig. 8): 75 Ω termination (R1+R2), gain 0.7, AC coupled by C1.
J2: RCA jack, hand-soldered (not in JLCPCB catalog).

Crystal 28.63636 MHz, fundamental, CL 20 pF:
 $C = 2(CL - C_s) - C_{pg} = 2(20 - 3) - 4 = 30 \text{ pF} \rightarrow 27 \text{ pF}$ (AN-1260)

Power: 3.3 V from Pi camera connector (pin 15). Two AP2112K-1.8 LDOs:
U2 $\rightarrow$ DVDD+MVDD (~84 mA typ). U3 $\rightarrow$ AVDD (~47 mA) and PVDD via FB1 (~12 mA).
UG-637 recommends separate regulated supplies for the analog/PLL group.

Reset / power-down / address
Overlay adv728x-m sets no reset/powerdown GPIO and does not enable cam1_reg.
so RESET and PWRDWN are pulled up on board; JP2/JP3 optionally route CAM_GPIO (Pi pin 11).
ALSB=1 $\rightarrow$ 7-bit addr 0x21 as in adv7282m-overlay.dts (reg = <0x21>). Close JP1 for 0x20.

MIPI CSI-2, 1 data lane + clock (216 Mbps interlaced / 432 Mbps I2P).
Route as 100 Ω differential (2 x 50 Ω loosely coupled per UG-637), no vias, GND plane under.

Pi 4 J3 (CAMERA) pinout, 1:1 with Camera Module v2.1:
1 GND 2 DON 3 D0P 4 GND 5 D1N(nc) 6 D1P(nc) 7 GND
8 CLKN 9 CLKP 10 GND 11 CAM_GPIO 12 nc 13 SCL0 14 SDA0 15 3V3
I2C on Pi 4 = GPIO44/45 (i2c-csi-dsi); pull-ups to 3V3 exist on the Pi.

Decoupling per UG-637: 100 nF + 10 nF at each supply pin, same side as U1, via under the 100 nF pad.

Sources: ADV7280 DS Rev.A, UG-637 Rev.A, AN-1260; see docs/requirements.md
Target: Raspberry Pi 4 CSI camera port, dtoverlay=adv728x-m,adv7280m=1

Sheet: /
File: adv7280m-csi2.kicad_sch

Title: CVBS to MIPI CSI-2 video decoder (ADV7280-M) for Raspberry Pi 4

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